

E-COMMERCE PLATFORM SEARCH FUNCTION

```
import java.util.ArrayList;
import java.util.Scanner;

class Product {
    int id;
    String name;
    double price;

    Product(int id, String name, double price) {
        this.id = id;
        this.name = name;
        this.price = price;
    }

    void display() {
        System.out.println("Product ID : " + id);
        System.out.println("Product Name : " + name);
        System.out.println("Price : ₹" + price);
        System.out.println();
    }
}

public class EcommerceSearch {

    public static void main(String[] args) {

        ArrayList<Product> products = new ArrayList<>();

        products.add(new Product(101, "Laptop", 65000));
        products.add(new Product(102, "Mobile", 25000));
        products.add(new Product(103, "Headphones", 2000));
```

```

products.add(new Product(104, "Keyboard", 1500));
products.add(new Product(105, "Mouse", 800));

Scanner sc = new Scanner(System.in);

System.out.print("Enter product name to search: ");
String search = sc.nextLine();

boolean found = false;

for (Product product : products) {

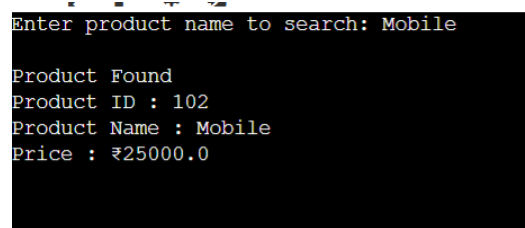
    if (product.name.equalsIgnoreCase(search)) {
        System.out.println("\nProduct Found");
        product.display();
        found = true;
        break;
    }
}

if (!found) {
    System.out.println("\nProduct not found.");
}

sc.close();
}
}

```

Output



```

Enter product name to search: Mobile

Product Found
Product ID : 102
Product Name : Mobile
Price : ₹25000.0

```

FINANCIAL AND FORECASTING

```
import java.util.Scanner;
```

```
public class FinancialForecast {
```

```
    public static double forecast(double revenue, double growthRate, int years) {
```

```
        if (years == 0) {  
            return revenue;  
        }
```

```
        return forecast(revenue * (1 + growthRate), growthRate, years - 1);  
    }
```

```
    public static void main(String[] args) {
```

```
        Scanner sc = new Scanner(System.in);
```

```
        System.out.print("Enter current revenue: ");  
        double revenue = sc.nextDouble();
```

```
        System.out.print("Enter annual growth rate (%): ");  
        double growth = sc.nextDouble();
```

```
        System.out.print("Enter number of years: ");  
        int years = sc.nextInt();
```

```
        double futureRevenue = forecast(revenue, growth / 100, years);
```

```
        System.out.printf("Forecasted Revenue after %d years = %.2f", years, futureRevenue);
```

```
        sc.close();  
    }  
}
```

Output

```
Enter current revenue: 1000  
Enter annual growth rate (%): 10  
Enter number of years: 5  
Forecasted Revenue after 5 years = 1610.51  
  
...Program finished with exit code 0  
Press ENTER to exit console.
```